

HAT-P-36b

R-0965

Benchmark run · workflow W8-benchmark-deep · record deep-bench_HATP-36_230131 · created 2026-07-29T16:35:55+00:00

BENCHMARK: ACCURATE — fit consistent with published values

Results

Mid-Transit Time (Tmid)	2459975.9582 +/- 0.0024 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.14 +/- 0.012
Transit depth (Rp/Rs)^2	1.96 +/- 0.33 [%]
Orbital Inclination (inc)	84.96 +/- 2.01
Airmass coefficient 1 (a1)	1.06 +/- 0.01
Airmass coefficient 2 (a2)	-0.053 +/- 0.048
Scatter in the residuals of the lightcurve fit is	0.99 %
Best Comparison Star	#1 - [277, 97]
Optimal Aperture	4.35
Optimal Annulus	11.31
Transit Duration (day)	0.0923 +/- 0.0055

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R■ fit vs published	0.14 ± 0.012 vs 0.126 (deviation 1.17σ)
Duration fit vs published	0.0923 d vs 0.0925 d (deviation 0.04σ)
Mid-time O–C	4.0 min
Depth SNR	11.7
Residual scatter	0.99 %

Light curve

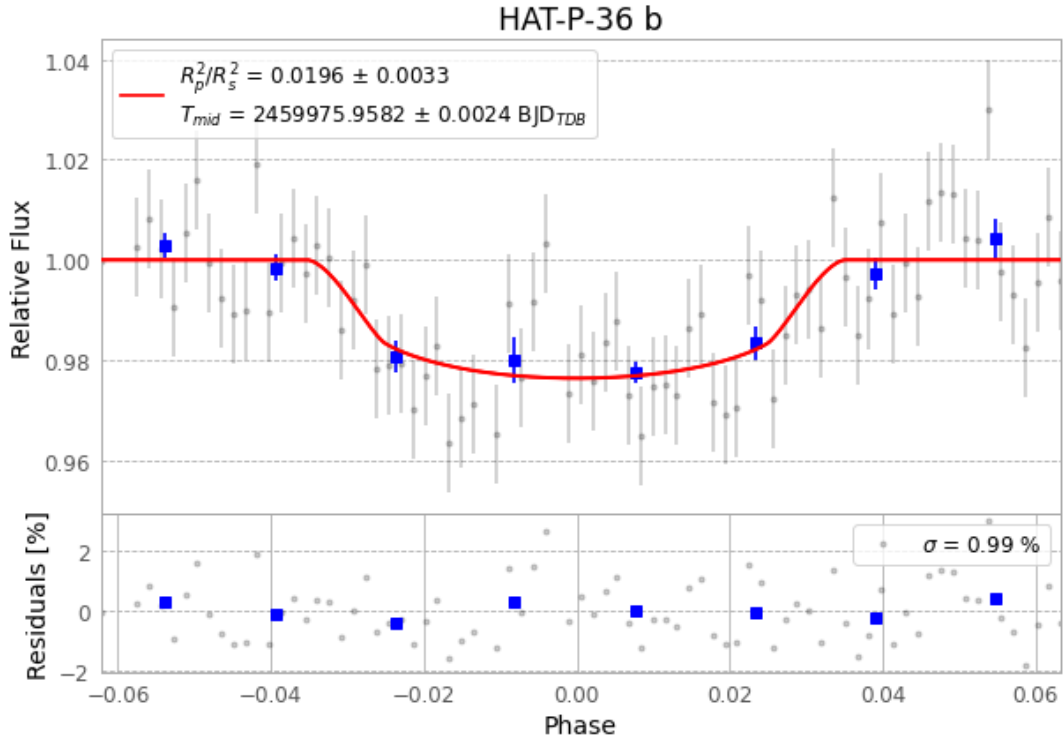


Figure 1 — FinalLightCurve_HAT-P-36 b_2023-01-31.png (SHA-256 9db21280af8cc48f...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [550, 159], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 e0b9c18f959ab39e...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit ead60b9

Data provenance

79 FITS frames (52 MB), HATP-36230131085413.FITS → HATP-36230131125415.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-HATP-36-230131.log (5e502dff080c...), FinalLightCurve_HAT-P-36 b_2023-01-31.png (9db21280af8c...), FinalLightCurve_HAT-P-36 b_2023-01-31.csv (99e85fedfe4d...)

Compute resources

Wall time 175.3 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-07-29 16:35Z.